

Cybersecurity Student Portfolio

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Information Technology Program

Wade College

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About Me

IT Student | Cybersecurity Focus

Motivated Information Technology student with a cybersecurity focus and hands-on experience in risk assessment, network security, secure architecture, cryptography, IoT systems, and e-commerce design.

Current Role: Information Technology Student, Wade College (Graduation: September 2026);
Cook Chill Specialist, City of Dallas, TX.

Career Goal: Cybersecurity professional focused on risk assessment, vulnerability management, network security architecture, and secure systems design.

Key Skills: Risk assessment, threat analysis, vulnerability management, AES-256, RSA, ECDSA, PKI, TLS, Zero Trust, HIPAA, NIST, IAM, MFA, MQTT, Raspberry Pi/Arduino, AWS/Azure, Python, Linux, PostgreSQL, Git, Wireshark.

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Project 01 - Northbridge Logistics Risk Assessment

Cybersecurity Risk Assessment & Defense Strategy | May 2026

Overview: 3 offices, 150 employees, 180 endpoints, 20 servers, cloud email, public customer portal, and remote access.

Threats: phishing/spear phishing (critical), ransomware (high), insider threat (high), man-in-the-middle (high), credential stuffing & brute force (high).

CIA Impact: confidentiality, integrity, and availability are all at risk.

Project 01 - Design & Strategy

Technical Controls: NGFW with deep packet inspection, EDR, MFA, automated patch management, email security gateway with sandboxing, WPA3 with VLAN segmentation.

Administrative Controls: security awareness training, AUP, password policy, data classification, incident response plan.

Physical Controls: badge/key-card access and CCTV surveillance.

Project 01 - Implementation

Priority 2 (30-60 days): deploy EDR across all endpoints.

Priority 3 (30-60 days): centralized automated patch management; close known CVEs within 30 days.

Priority 4 (60-90 days): deploy WAF and conduct external penetration testing.

Priority 5 (60-90 days): mandatory phishing-recognition and incident-reporting training.

Project 01 - Results & Reflection

Key Learning: cybersecurity requires continuous improvement across technology, processes, and people.

Defense-in-depth works through overlapping controls.

The MGM Resorts case study reinforced that social engineering can bypass technical controls.

Project 02 - Cryptography & Network Security

SecureHealth Systems handles PHI across internal EHR servers, cloud services, and partner hospital connections.

Attack vectors: eavesdropping, MITM, ransomware, insider threats, and DoS.

Objectives: confidentiality with AES-256, integrity with HMAC and digital signatures, authenticity and non-repudiation for clinical decisions.

Project 02 - Design & Strategy

AES-256-GCM for data in transit.

RSA-2048 for legacy compatibility; ECDSA P-256 for digital signatures; ECDH for ephemeral session keys.

SHA-256 for signatures/HMAC; SHA-3-256 for root CA fingerprinting.

HMAC-SHA-256 on API payloads and database records; ECDSA signatures on clinical records.

Three-tier PKI with offline Root CA and online Intermediate CA; private keys stored in FIPS 140-3 HSM.

Project 02 - Implementation

Key lifecycle: HSM-based generation, X.509 v3 distribution, HSM storage, 90-day rotation, CRL and OCSP stapling.

DMZ: patient portal with WAF and TLS 1.3; no direct database access.

Internal: EHR servers, AES-256-CBC at rest + HMAC-SHA-256; mutual TLS between services.

Cloud: client-side AES-256-GCM before upload.

Partners: site-to-site IPsec VPN with IKEv2; mutual TLS for HL7 FHIR API calls.

Project 02 - Results & Reflection

HIPAA verification: PHI encrypted in transit and at rest, audit controls, integrity controls.

Key Insight: strong cryptography depends on disciplined key generation, distribution, storage, rotation, and revocation.

Project 03 - IoT Smart Home Monitoring

Objective: prototype system using temperature, humidity, and air-quality sensors with Raspberry Pi/Arduino and cloud dashboard.

Challenge: integrate heterogeneous sensors, transmit reliably over wireless networks, and provide real-time and historical visualization.

Targets: 95% sensor accuracy, 10-second refresh, 95%+ uptime, alerts through a user-friendly interface.

Project 03 - Design & Strategy

MQTT for lightweight publish-subscribe communication.

Cloud broker: AWS IoT Core or Blynk; time-series storage for historical analysis.

Dashboard: real-time charts/gauges and threshold alerts by email/SMS/push.

Security: MQTT over SSL/TLS, API tokens, encrypted cloud storage.

Project 03 - Implementation

Phase 1: planning, procurement, GPIO wiring, Raspberry Pi OS, Python libraries.

Phase 2: Python firmware, MQTT publishing, cloud setup, connectivity testing.

Phase 3: dashboard, calibration, alert rules, data persistence.

Phase 4: 7-day reliability test, accuracy verification, Wi-Fi dropout recovery, dashboard load testing.

Project 03 - Results & Reflection

Deliverables: functional prototype, 10-second collection, 95%+ accuracy, 30-day historical trends, alerts, technical documentation.

Key Learnings: MQTT is efficient; calibration is essential; cloud dashboards and network resilience matter.

Future: early procurement, secondary Wi-Fi AP, offline buffering for cloud outages.

Project 04 - E-Commerce Storefront: Makeda

Opportunity: authentic Ethiopian and African fashion is underrepresented in mainstream e-commerce.

Challenge: establish cultural authenticity and brand trust with no initial brand recognition.

Scope: Shopify storefront, product catalog, SEO strategy, cohesive brand identity, payments and shipping.

Project 04 - Design & Strategy

Store architecture: homepage, curated collections, About Us, Contact, mobile-first design.

Brand: Makeda - The Queen of Sheba; elegant minimalism with culturally inspired accents.

SEO keywords: authentic Ethiopian fashion, African clothing online, traditional habesha clothing, diaspora style, sustainable African fashion.

Project 04 - Implementation

Platform: Shopify with Horizon theme.

Store URL: queen-of-sheba.myshopify.com

Catalog: 15+ SKUs across traditional and contemporary African fashion.

Payments/logistics: Shopify Payments, Stripe/PayPal, diaspora shipping zones, GDPR-conscious handling.

Apps: Judge.me reviews, newsletter capture, abandoned-cart recovery.

Project 04 - Results & Reflection

Strengths: unique niche and strong cultural identity.

Weaknesses: new store and limited launch catalog.

Opportunities: growing diaspora demand and social-media reach.

Threats: larger competitors and rising international shipping costs.

Strategic Insight: success connects authentic storytelling, mobile-first UX, and discoverability through SEO/social media.

Get in Touch

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